BC3010

Bioinformatics

Note: Approval needed from Academic School/Department before you can take this module.

Subject Area: BC - Biochemistry and Cell Biology

Credit Weighting: 5

The notional student workload for 5 credits is 125 hours

Semester(s)

Semester 2 (Jan-May)

No. of Students

Min 30, Max 100

Pre-requisite(s)

BC2001 - Biomolecules **AND**
ML2001 - Introductory Molecular Biology **OR**
ML2901 - Introductory Molecular Biology

Co-requisite(s)

None

Teaching Methods

- 18 x 1 Hr(s) Lectures
- 5 x 2 Hr(s) Practicals/Laboratories

Module Co-ordinator

Prof Pavel Baranov - School of Biochemistry and Cell Biology

Lecturer(s)

- Prof Pavel Baranov - School of Biochemistry and Cell Biology

Guest Lecturer(s)

None

Module Goal

Introduction to Bioinformatics.

Module Content

Probability basics. Conditional probabilities. Bayes Theorem. Bit. Uncertainty, Entropy and Information. Information in nucleotide and protein sequences. Information flow during gene expression. Genetic Code and translation as a noisy communication channel. Genome annotations. RNA secondary structure. Mutual Information. Sequence Logos. Relative Entropy (Kullback-Leibler divergence). Comparative sequence analysis. Hamming and Levenstein distances. Weight matrices. Pairwise sequence alignment. Dynamic programming. Progressive multiple alignment. Heuristics methods (BLAST). Molecular Evolution. Phylogenetic Analysis. Trees. Phenetics and Cladistics. Databases. Genome Browsers and other Internet Resources. Differential gene expression analysis. Machine learning techniques, Hidden Markov Models, PSI-BLAST.

Learning Outcomes

On successful completion of this module, students should be able to:

1. Quantitatively express information in biological sequence patterns and visualize information content using Sequence Logos.
2. Explain how biological molecules store, replicate and transform information and how redundancy contributes to the accuracy of these processes.
3. Use compensatory mutations to identify evolutionary conserved RNA secondary structures.
4. Recall basic features of eukaryotic and prokaryotic protein coding genes.
5. Find optimal pairwise alignment using Needleman-Wunsch algorithm.
6. Differentiate heuristics approaches from optimal solutions.
7. Perform simple comparative sequence analysis (frequency and compositional analysis) and sequence similarity searches (BLAST).
8. Discriminate Phenetics and Cladistics.
9. Use online databases, genome browser and other internet resources.

Examination and Assessment

Total Marks 100

- Written Examination 70 Marks - Semester 2 Written Exam - Summer
  - Paper 1: 1.5hr paper - MCQ (70 Marks)
- Continuous Assessment 30 Marks
  - Attendance & Participation - Computer Laboratory Project (20 Marks)
  - Quiz(zes) - MCQ (10 Marks)

Penalties (for late submission of Course/Project Work etc.)

Work which is submitted late shall be assigned a mark of zero (or a Fail Judgement in the case of Pass/Fail modules)

Pass Standard

40%

Special Requirements for Passing Module

None

Supplemental Examination and Assessment

- Written Examination - Autumn
  - Paper 1: 1.5 hour paper - MCQ
- Continuous Assessment

- The mark for continuous assessment is carried forward
    - The marks attained in the MCQ Continuous Assessment and the Computer Laboratory Project are carried forward.